

Computer Networks Project



Department of
Computer Science
Faculty of ICT, BUITEMS

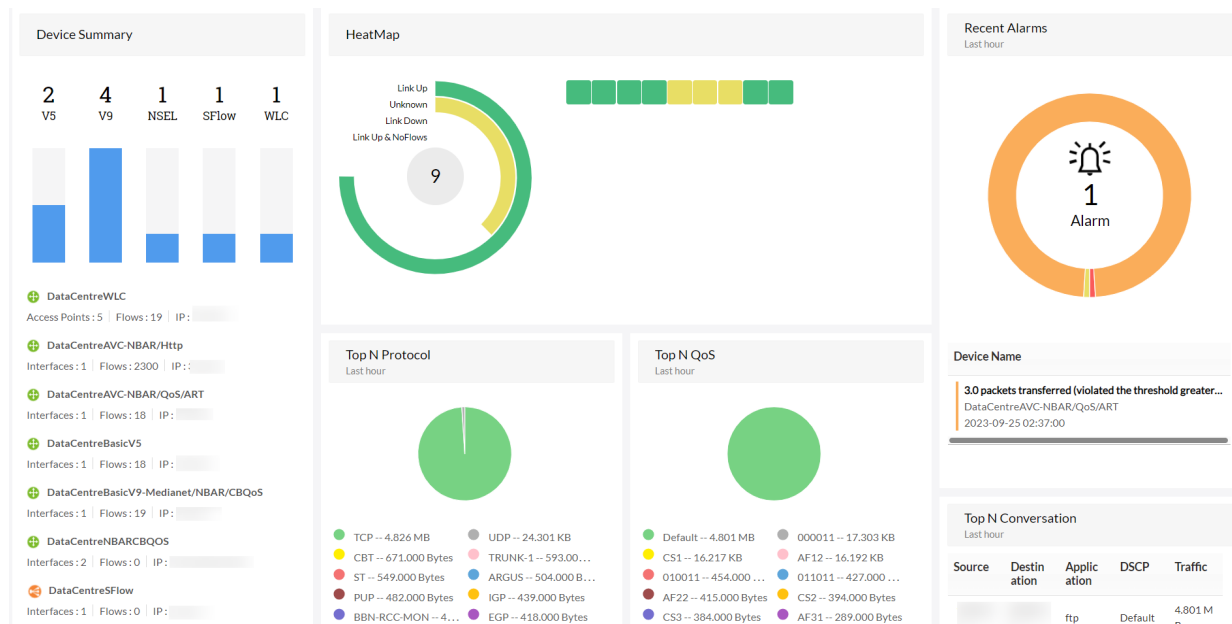
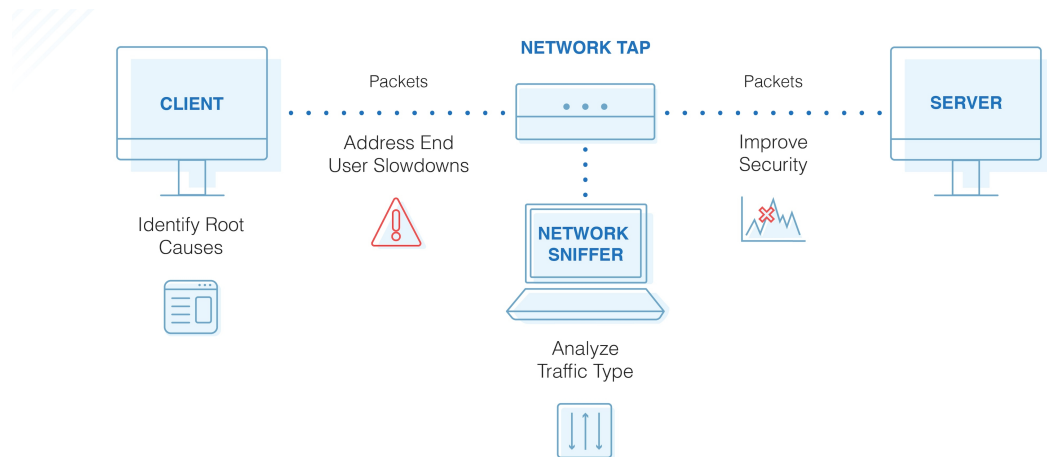
BUITEMS Faculty of Information and
Communication Technology



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Title

Packet Sniffer and Network Traffic Analyzer using Python



Course Learning Outcome			
Sr. No	CLO Involved	Learning Outcomes / Justification	Points
1	CLO-4	Design and Implement networking model or a routing technique, write a report and present your work.	12

Course Information							
Course Code	C	S	3	0	5	Course Title	Computer Networks
Semester	3rd				Total Marks	12	
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1 Project Description

This project focuses on the development of a simple packet sniffing and network traffic analysis tool using Python. A packet sniffer is a program that captures network packets moving through a network interface. By analyzing these packets, students can understand how data travels across a network and how different protocols are used for communication.

The project will help students practically understand important Computer Networks concepts such as IP addresses, MAC addresses, ports, protocols, packet size, TCP, UDP, ICMP, DNS, HTTP, HTTPS, and network traffic behavior.

Students will create a Python-based tool that can capture live network packets or read packets from a saved packet capture file. The tool should extract useful information from each packet and display a simple traffic summary. The system may also generate basic statistics, such as the number of TCP packets, UDP packets, ICMP packets, most active IP addresses, and traffic distribution.

This project is suitable for second-semester students because it is practical, easy to understand, and directly connected with Computer Networks.

2 Project Motivation

Computer networks are used everywhere, including homes, universities, offices, banks, hospitals, and companies. Every time a user opens a website, sends a message, downloads a file, or watches a video, packets are transmitted across the network.

Understanding packets is one of the most important parts of Computer Networks. Packet sniffing allows students to observe real network communication and understand how different protocols work in practice.

This project will give students hands-on experience with network traffic analysis instead of only studying theoretical concepts.

3 Main Objectives

The main objectives of this project are:

- To understand how network packets are transmitted across a network.
- To capture and analyze network packets using Python.
- To identify basic protocol types such as TCP, UDP, ICMP, DNS, HTTP, and HTTPS.
- To extract useful information such as source IP, destination IP, source port, destination port, protocol, and packet size.

- To count and summarize captured packets.
- To generate simple traffic statistics and visualizations.
- To save captured packet information into a CSV file.
- To prepare a simple 6-page project report.
- To present the working project in class.

4 Group Requirement

Students must work in groups.

- Each group must consist of **three students**.
- All group members must contribute to the project.
- Work should be divided properly among group members.
- Each student should understand the complete project, not only one part.
- During the presentation, any group member may be asked to explain the code, report, or project working.

5 Tools and Technologies

Students may use the following tools and technologies:

- Python
- Scapy or PyShark
- Wireshark
- Pandas
- Matplotlib
- Jupyter Notebook or Python scripts
- GitHub
- Microsoft Word, Google Docs, or Overleaf for report writing

6 Recommended Python Libraries

Students may use the following Python libraries:

- `scapy` for packet capturing and packet analysis
- `pyshark` as an alternative packet analysis library
- `pandas` for storing and processing packet data
- `matplotlib` for creating graphs
- `csv` for saving packet records
- `datetime` for recording packet capture time

7 System Requirements

The project can be implemented on:

- Windows, Linux, or macOS
- Python 3.8 or above
- Wireshark installed for packet verification
- Administrator/root permission may be required for live packet capturing
- Internet connection or local network connection for testing

8 Project Features

The project should include the following features:

8.1 Basic Features

- Capture live network packets.
- Display source IP address.
- Display destination IP address.
- Display source port number where available.
- Display destination port number where available.
- Identify protocol type such as TCP, UDP, ICMP, or others.

- Display packet size.
- Display packet capture time.
- Save captured packet data into a CSV file.

8.2 Traffic Analysis Features

- Count total number of captured packets.
- Count TCP packets.
- Count UDP packets.
- Count ICMP packets.
- Count other packets.
- Show top source IP addresses.
- Show top destination IP addresses.
- Show most commonly used destination ports.
- Generate a simple protocol distribution graph.

8.3 Optional Advanced Features

These features are optional. Students can implement them for better marks.

- Read and analyze packets from a .pcap file.
- Detect DNS traffic.
- Detect HTTP or HTTPS traffic based on port numbers.
- Create a simple menu-based command-line interface.
- Create a simple graphical dashboard.
- Add filtering by protocol, IP address, or port number.
- Export analysis summary as a text file or PDF.

9 Proposed Methodology

The project should be completed in the following steps:

9.1 Step 1: Study Basic Network Concepts

Students should first revise the following concepts:

- IP address
- MAC address
- Ports
- TCP protocol
- UDP protocol
- ICMP protocol
- DNS protocol
- HTTP and HTTPS
- Packet structure

9.2 Step 2: Install Required Tools

Students should install Python, Wireshark, and required Python libraries. They should test whether packet capturing is working on their system.

9.3 Step 3: Capture Network Packets

The program should capture packets from the network interface. Students may either capture live packets or use a saved packet capture file for testing.

9.4 Step 4: Extract Packet Information

For each captured packet, the program should extract useful information such as:

- Time of capture
- Source IP address
- Destination IP address
- Source port
- Destination port
- Protocol type
- Packet length

9.5 Step 5: Store Packet Data

The extracted packet information should be stored in a structured format. Students may save the data in a CSV file.

9.6 Step 6: Analyze the Traffic

The program should analyze the captured packets and generate a summary. The summary may include protocol counts, top IP addresses, top ports, and total captured packets.

9.7 Step 7: Visualize the Results

Students should create at least one graph, such as:

- Protocol distribution bar chart
- Protocol distribution pie chart
- Top source IP addresses graph
- Top destination ports graph

9.8 Step 8: Prepare Final Report

Students must write a simple 6-page report explaining the project, implementation, results, and conclusion.

9.9 Step 9: Present the Project

Each group must present the working project before the final evaluation.

10 Expected Output

The final project should produce outputs such as:

- A list of captured packets.
- Source and destination IP addresses.
- Source and destination ports.
- Protocol type of each packet.
- Packet size.
- CSV file containing captured packet details.

- Summary of packet counts.
- Graph showing protocol distribution.

11 GitHub Repository Requirement

Each group must create a GitHub repository for the project. The repository should be properly organized and easy to understand.

The recommended repository structure is:

```
packet-sniffer-network-analyzer/
```

```
README.md
```

```
requirements.txt
```

```
src/
```

```
    packet_sniffer.py
```

```
    analyzer.py
```

```
    visualizer.py
```

```
data/
```

```
    captured_packets.csv
```

```
figures/
```

```
    protocol_distribution.png
```

```
    top_ips.png
```

```
report/
```

```
    final_report.pdf
```

```
screenshots/
```

```
    output_screenshot.png
```

The README.md file should include:

- Project title
- Group member names
- Short project description
- Tools and libraries used

- Installation instructions
- How to run the project
- Sample output
- Screenshots

12 Report Requirement

Each group must submit a simple project report of approximately **6 pages**. The report should be written in simple document style. It does not need to follow IEEE format.

The report should include the following sections:

12.1 Title Page

The title page should include:

- Project title
- Course name
- Group member names
- Instructor name
- Department name
- Submission date

12.2 Introduction

This section should explain what packet sniffing is and why network traffic analysis is important.

12.3 Objectives

This section should list the main goals of the project.

12.4 Tools and Technologies

This section should mention the tools and libraries used in the project.

12.5 Methodology

This section should explain how the project was implemented step by step.

12.6 Implementation

This section should explain the main code modules, packet capturing method, packet analysis method, and data storage method.

12.7 Results and Screenshots

This section should include screenshots of the working project, captured packet output, CSV file, and graphs.

12.8 Conclusion

This section should explain what students learned from the project and how the project helped them understand Computer Networks.

12.9 References

Students should mention any websites, documentation, tutorials, or books used for help.

13 Presentation Requirement

Each group must present the project in class.

The presentation should include:

- Project title and group member names
- Problem introduction
- Objectives
- Tools and technologies used
- Working of the packet sniffer
- Code explanation
- Results and screenshots
- Live demonstration
- Conclusion

All group members must participate in the presentation. Each student should be able to explain their contribution.

14 Must Requirements and Rules

Students must follow the requirements given below:

- Each project must be completed by a group of **three students**.
- All group members must contribute equally.
- The project must be implemented using Python.
- The project must capture or analyze network packets.
- The project must show packet details such as source IP, destination IP, protocol, and packet size.
- The project must save results in a CSV file.
- The project must include at least one graph or chart.
- The project must include a GitHub repository.
- The project must include a 6-page report.
- The project must be presented in class.
- Students must understand their own code.
- Copied projects without understanding will not be accepted.
- Fake screenshots or fake results will not be accepted.
- Late submission may result in marks deduction.

15 Learning Outcomes

After completing this project, students should be able to:

- Understand how packets travel in a network.
- Identify common network protocols.
- Capture and analyze packets using Python.
- Use Wireshark for packet verification.
- Store and process packet data.
- Generate simple network traffic statistics.

- Visualize network traffic using graphs.
- Work as a team on a technical project.
- Present a working Computer Networks project.

16 Conclusion

This project is a practical introduction to network packet analysis. It allows students to observe real network communication and understand the role of protocols, IP addresses, ports, and packet structures.

By completing this project, students will gain hands-on experience in Computer Networks and Python programming. The final output should include a working packet sniffer, traffic analysis results, a GitHub repository, a 6-page report, and a class presentation.